Maternal Smoking and its Impact on Infant Birthweight

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Econ 310: Introduction to Econometrics

Introduction

Smoking while pregnant can have many negative effects on the health of the infant as well as the mother. According to Office of the Assistant Secretary for Health twenty two percent of women self-identified as smokers. This poses a problem not just for the infant but as well as for the mother as complications could arise due to health conditions during birth. In the data set provided by Professor Douglas Almond, Kenneth Chay, and David Lee we are given information on the variables that could affect the birthweight of infants. I want to focus on how smoking directly affects the weight of infants, but we first need to find out just how much other variables might affect the results. These variables can either negatively or positively affect the outcome.
The following analysis helps us answer the question of, "Does smoking during pregnancy affect the birthweight of newborns?"

We want to set up our hypothesis to properly define the scope of our investigation. When testing the relationships between variables, we want to make sure that we have a well-defined guide. As stated earlier, we want to find out whether smoking affects newborns' birthweight or not. We can set our null hypothesis(H0) that suggests no effect or difference or in other words, smoking is not a cause in the birthweight of newborns. The alternative hypothesis(H1) would suggest that there is an effect causing a change or alternatively smoking does cause a change in newborns birthweight. We can set the null hypothesis as H0: $\mu 1 = \mu 2$ where $\mu 1$ is the mean birthweight of babies born to non-smoking mothers and $\mu 2$ represents the mean birthweight of babies born to smoking mothers. The alternative hypothesis can be set as H1: $\mu 1 \neq \mu 2$ where $\neq$ indicates we we're interested in the birthweight regardless of the direction.

The significance level (α) should be set to 0.05 as this means there is a 5 % chance of making a type I error which is the probability of rejecting the null hypothesis when it is true. We will go more in depth into the significance level when discussing the findings from the data. To get started, we want to focus on the data that is being provided and how they affect each other. As I mentioned before, we want to find out whether the variables affect the birthweight by being statistically significant. This could mean that we might need to drop some variables to make our model fit correctly.

<u>Data Description: Table of summary stats</u>

Table 1.

Variable	Obs.	Mean	Std. Dev.	Max.	Min.
Nprevist	30000	11.1	3.67	35.00	0
Alcohol	30000	0.02	0.14	1.00	0
Tripre1	30000	0.80	0.4	1.00	0
Tripre2	30000	0.15	0.36	1.00	0
Tripre3	30000	0.03	0.18	1.00	0
Tripre0	30000	0.01	0.10	1.00	0
Birthweight	30000	3382.93	592.16	5755.00	425.00
Smoker	30000	0.19	0.40	1.00	0
Unmarried	30000	0.22	0.41	1.00	0
Education	30000	12.91	2.17	17.00	0
Age	30000	26.89	5.40	44.00	14.0
Drinks	30000	0.06	0.69	21.00	30

Variable definitions: Nprevist = total number of prenatal visits; Alcohol = binary variable (1 if mother drank during pregnancy, 0 otherwise); Tripre1 = binary variable (1 if 1st prenatal care visit in 1st trimester, 0 otherwise); Tripre2 = binary variable (1 if 1st prenatal care visit in 2nd trimester, 0 otherwise); Tripre3 = binary variable (1 if 1st prenatal care visit in 3rd trimester, 0 otherwise); Birthweight = birthweight of infant (in grams); Smoker = binary variable (1 if mother smoked during pregnancy, 0 otherwise); Unmarried = binary variable (1 if mother is unmarried, 0 otherwise); Education = years of educational attainment (more than 16 years coded as 17); Age = age; Drinks = binary variable (1 if mother drank alcohol during pregnancy, 0 otherwise). All observations conducted in Pennsylvania in 1989.

This table gives us insight into the data set which is comprised of 30,0000 observations. Notably, we can point out numbers such as 11.1 which is the average number of visits during pregnancy with the average birthweight being 3382.93 grams. Another notable figure in this data set is alcohol with an average of 0.02 which is relatively low. Demographic aspects such as age with an average of 26.89 years and education with an average of 12.91 years are also covered in this data set.

This data was collected to show the relationship between smoking and birthweight of a newborn, but it also offers a glimpse into in the lifestyle of the population that was sampled which in this case was pregnant women. All the variables together make this data set a valuable source to help us understand the relationship mothers and their children have within a sizeable relationship.

In this data set we're dealing with multiple variables meaning one scatterplot might not be enough to show the relationships these variables share with each other. In this case, we want to see how the independent variables affect the dependent variable. Before we go into creating scatter plots for each variable, it would be best to find which variables have the strongest correlation to the dependent variable birthweight.

We can use the following formula to find the correlation coefficient for the variables using,

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \cdot \sum(Y_i - \bar{Y})^2}}\ .$$

Table 2.

Variable	Correlation
Nprevist	0.23
Alcohol	-0.03
Tripre1	0.11
Tripre2	-0.07
Tripre3	-0.05
Tripre0	-0.12
Birthweight	1.0
Smoker	-0.17
Unmarried	-0.20
Education	0.11
Age	0.08
Drinks	-0.03

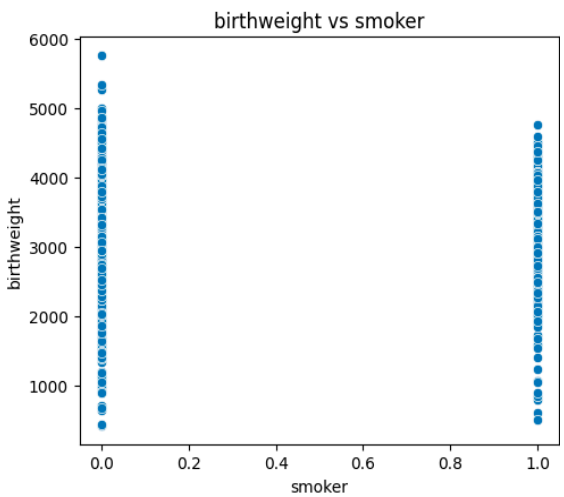
The correlation column tells us how closely related to the dependent variable is to the rest of the variables. It would be preferable to be as close to 1 or -1. In this case, the variables might have a negative relationship with the variable which could explain the negative correlation.

When analyzing the data, we can see that some of the variables have more of a correlation than others. If we look at the variable Birthweight, we can see that it has a correlation of 1 and this is due to it being the target variable so it would make sense for the correlation to be a 1. Furthermore, some of the variables seem to have a weak correlation to birthweight which gives us a reason to remove them before we conduct a linear regression.

We can drop alcohol, tripre2, tripre3, age and drinks from the data set to conduct the linear regression. We can see alcohol has a correlation of -0.03, tripre2 a correlation of -0.07, tripre3 a correlation of -0.05, age a correlation of 0.08, and drinks a correlation of -0.03. This does not mean that the variables do not have a relationship with each other, they just don't have a linear relationship together. If we compare our other variables to the ones I listed, we can see they have a stronger relationship whether it be negative or positive.

Background of your model: Scatter plot of Data

Graphs give us the ability to further analyze the data we are working with by examining the data visually.



When we focus on one of the variables such as smoker, we notice that the data is very concentrated. This is partly due to smoker being a binary variable meaning that it can only have two outcomes.

This data makes it difficult to visualize how the data is spread but we can still analyze it. At 0 for nonsmoker, we see the data is spread out in comparison to 1 or smoker. We can also see we have some outliers in the graph for nonsmokers which might change the results of the regression.

Regression Formula

To form the regression formula, we need to set the variables we want to use. The general linear regression is in this form:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}X_1$$

The independent variable is represented by $\hat{Y}$, $\widehat{\beta_0}$ represents the intercept, $\widehat{\beta_1}$ represents the slope coefficient, and X_1 represents the independent variable. This data set has multiple variables which is why we need to use a similar formula which is only an extension of our general linear regression form:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}X_1 + \cdots + \widehat{\beta_n}X_n$$

We are now able to add as many variables as we see fit although we need to make sure the number of variables and the number of observations makes sense. Many variables and a small number of observations would not be a good fit for a model.

To find the coefficients for the formula, we need to calculate them with the formula:

$$\widehat{\beta_1} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

To know the value of the dependent variable when the independent variables are 0, the intercept should be calculated. We can calculate the intercept by using the following formula:

$$\widehat{\beta_0} = \bar{Y} - \beta_1 \bar{X}$$

When the values are calculated we get the following results shown in table 3.

Table 3.

Variable	Coefficient
Cons	3185.58
nprevist	30.78
Tripre1	-75.81
Tripre0	-303.54
Smoker	-177.48
Unmarried	-197.96
Education	0.17

As we input the coefficients, our formula now begins to take shape. Notice the negative coefficients, it is very important to consider whether the coefficient is negative or positive as it captures the relationship between the dependent variable and the independent variables. In this data set our coefficients are telling us how the weight might move such as nprevist telling us that as the constant can now tell us where our regression line needs to be plotted. The intercept also tells us how much an infant would weigh if all the independent variables were set to 0.

If we put our variables, coefficients, and constant or intercept we can now form the regression formula which in this case would be:

$\widehat{Birthweight}$= 3185.58 + 30.79 x nprevist – 75.81 x tripre1 – 303.54 x tripre0 – 177.48 x smoker – 197.96 x unmarried + 0.17 x educ

Show results in table

Table 4.
Dependent Variable: Birthweight

	Coefficient	Std. Error	t	P-Value	Lower	Upper
Cons	3185.58	74.59	42.70	0	3039.32	3331.83
nprevist	30.78	3.32	9.27	0	24.27	37.30
Tripre1	-75.81	30.92	-2.45	0.014	-136.44	-15.19
Tripre0	-303.54	109.14	-2.78	0.005	-517.53	89.55
Smoker	-177.48	27.24	-6.52	0	-230.9	-124.07
Unmarried	-197.96	27.52	-7.19	0	-251.93	-143.99
Education	0.17	5.23	0.03	0.973	-10.08	10.43
R-Squared	0.093					
Adj. R-Squared	0.091					
F-Statistic	50.88					

Discuss results in table

From table 4 we can see the coefficients, standard errors, t-stats, p-values, as well as the lower and upper confidence intervals. We can see the standard errors are small relative to the coefficients letting us know our variables are reliable. The only variable that seems unreliable is education. If we take a closer look at the variable, we see that the P-value is 0.973 which does not fit with the significance level. We can remove this variable which means we need to develop a new model without the education variable.

Table 5.
Dependent Variable: Birthweight

	Coefficient	Std. Error	t	P-Value	Lower	Upper
Cons	3185.58	74.59	42.70	0	3039.32	3331.83
nprevist	30.78	3.32	9.27	0	24.27	37.30
Tripre1	-75.81	30.92	-2.45	0.014	-136.44	-15.19
Tripre0	-303.54	109.14	-2.78	0.005	-517.53	89.55
Smoker	-177.48	27.24	-6.52	0	-230.9	-124.07
Unmarried	-197.96	27.52	-7.19	0	-251.93	-143.99
R-Squared	0.093					
Adj. R-Squared	0.091					
F-Statistic	61.08					

The f-statistic emphasizes the models fit for this data although the R-Squared does seem to be relatively low as it sits at 9.3%. This could be due to outliers in the data set or the number of variables being too low. Now we can see in table 5 most of the p-values are 0 or near 0. This makes all our variables statistically significant but also gives us the chance to reject the null hypothesis stated earlier which was that smoking did not cause a change in birthweight. This means we have evidence to reject the alternative hypothesis were it was stated that smoking can affect the birthweight of an infant. This new table also changes the linear regression formula which would be:

$\widehat{Birthweight}$= 3185.58 + 30.79 x nprevist – 75.81 x tripre1 – 303.54 x tripre0 – 177.48 x smoker – 197.96 x unmarried

<u>Conclusion</u>

The model developed showed evidence that smoking was a probable cause of change in birthweight specifically lowering birth weight. The significance of the variable coefficients, supported by the low p-values showed the effects of the independent variables. The analysis shows the implications for public health initiatives for child wellbeing. The study for this data was conducted in 1989 and as of 2006 women's smoking has reduced to 18% according to the national center for biotechnology information. It is important to continue monitoring these statistics and collecting data to prevent further problems.

These findings emphasize the need for awareness and educational programs to reduce smoking during pregnancy and at times before pregnancy. While these results provide evidence for a reduction of weight for infants due to smoking, other variables not listed may be mor likely to cause an affect. Nevertheless, this investigation provides valuable insight into the broader understanding of factors that influence birthweight and highlights the importance of addressing maternal smoking for the health of all parties involved.

Citations:

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